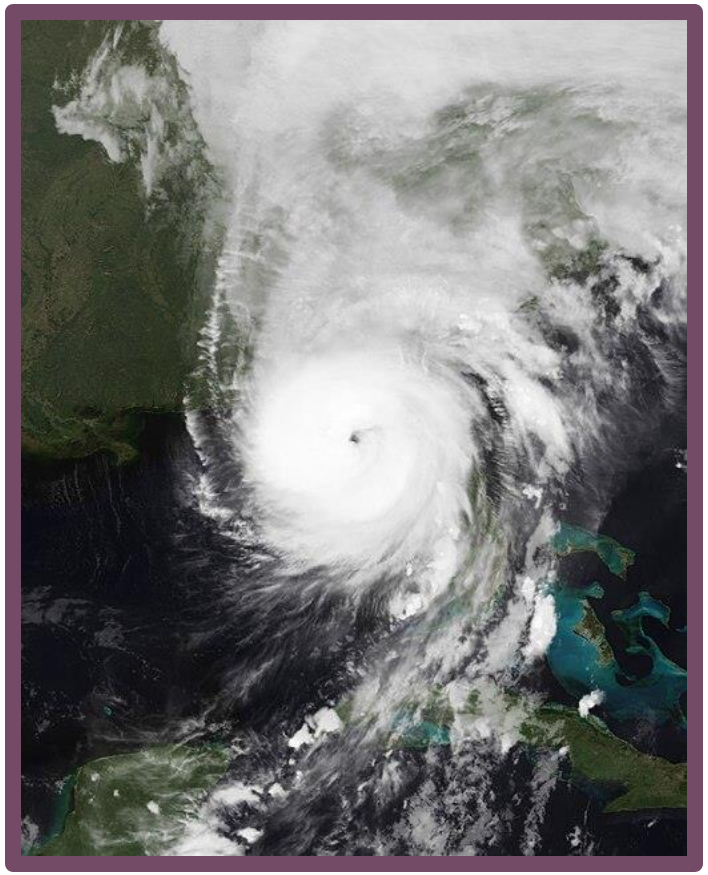
OPERATIONAL RISK



Hurricane Helene
September 2024

ENP-0112

2/18/2026

Dr. Divya C. Chandra

MITIGATIONS

Individual

- Selection and training (Knowledge, Skills, and Abilities)
- Motivation, arousal (optimize attention)

CRM
Monitoring
Fatigue

Organization

- Policies and procedures (e.g., checklists)
- Safety culture

System Design

Displays and Alerts

- Controls (soft or hard), “Usability” (e.g., visibility of system state, interaction with data)
- Automated systems? (take the operator out of the system)

Operational Risk

- Variation across each operation (e.g., across flights)
- Work as imagined (theory) vs. work as done (reality)

TOPICS

- Operations
  - Threat and error framework, sources of risk
- Safety 1 and Safety 2
  - Human error vs. human contribution to safety
- Analysis of a sample ASRS event
  - Limitations
  - What was the pilot's contribution to safety?

OPERATIONS

THREAT AND ERROR MANAGEMENT (TEM) FRAMEWORK

- **Threats and errors** are part of everyday aviation operations that must be managed by aviation professionals.
- **Undesired states** carry the potential for unsafe outcomes.
- **Undesired state management** largely represents the last opportunity to avoid an unsafe outcome and thus maintain safety margins in aviation operations.
- TEM is analogous to "defensive driving." It emphasizes techniques that minimize risks related to safety.

https://skybrary.aero/articles/threat-and-error-management-tem
Merritt & Klinect (2006). *Defensive Flying for Pilots: An Introduction to Threat and Error Management. https://skybrary.aero/sites/default/files/bookshelf/1982.pdf*

ORIGINS OF TEM

Do the concepts taught in training transfer to normal everyday flight operations?

- Assess through Line Operations Safety Audits (LOSA), now the LOSA Collaborative. (Originally a partnership between University of Texas researchers and airlines.)
- Designed to assess the effectiveness of CRM training, expanded to address error and its management.
- FAA endorses LOSA as a voluntary safety program (FAA AC 120-90)

THREAT DEFINITION

- Threats are events or errors that
  - Occur outside the influence of the flight crew (i.e., not caused by the crew)
  - Increase the operational complexity of a flight, and
  - Require crew attention and management if safety margins are to be maintained
- Examples
  - High terrain, icing conditions, aircraft malfunction, other people's errors (e.g., dispatcher error)
  - Two categories
    - **Environmental threats** are outside the control of the airline
    - **Airline threats** originate from the flight operations (e.g., aircraft malfunctions)
  - All are independent of flight crew, but must be managed by the crew

ERROR DEFINITION

- Errors are defined as flight crew actions or inactions that
  - Lead to a deviation from crew or organizational intentions or expectations
  - Reduce safety margins; and
  - Increase the probability of adverse operational events on the ground or during flight
- Examples – Three Categories
  - **Aircraft handling,** e.g., errors related to manual flying, or managing automated systems
  - **Communication errors**, e.g., related to ATC communications, or pilot-to-pilot comms
  - **Procedural errors,** e.g., deviations from regulations or standard operating procedures)

MANAGING RISK: OPERATIONAL COMPLEXITY IN FLIGHT OPERATIONS

Sources of Operational Complexity
- ATC interventions
- Aircraft equipment or performance
- Crew factors
- Operator factors
- Environment factors

ATC Intervention (such as)
- (Late) route amendments
- Unpublished restrictions
- Vectors

Operator Factors
- Independence vs. dependence on Dispatch
- Clarity and consistency of pilot roles in reviewing route

Aircraft Factors
- Lack or unreliability of automated systems
- Performance characteristics

Environment Factors
- Terrain
- Traffic
- Weather
- Prohibited airspace

Crew Factors
- Expectations (confirmation bias)
- Fatigue
- Communication style
- Distractions
- Local area familiarity
- Familiarity with different types of routes

Chandra, D.C., & Markunas, R. (2017). “Line Pilot Perspectives on Complexity of Terminal Instrument Flight Procedures,” DOT-VNTSC-FAA-17-06. U.S. DOT Volpe National Transportation Systems Center.

SAFETY 1 AND SAFETY 2

SAFETY 1 AND SAFETY 2

- Safety 1 examines what went wrong
  - Safety = absence of accidents or an acceptable level of risk
  - Analyze accident/incident investigations, look for remedies
    - Humans, organizations, and technology are a source of error, failure, and malfunctions
    - Accidents/incidents are relatively rare

- In contrast, Safety 2 examines everything that what went right
  - Risks are always being managed, human variability is necessary and helpful in many situations.
  - The vast majority of flight operations are safe. Study these.

MORE ON SAFETY 1 AND SAFETY 2

- Safety 1 and Safety 2 are complementary views. Need to consider both.

- In Safety 1, when things go right, it's because the work went as designed/imagined. Reinforce compliance and eliminate variability. Systems are decomposable into components. Root cause can be identified.

- In reality systems are complex and inter-dependent. Trying to constrain performance can make it more difficult to achieve the desired outcomes and be counterproductive. Human flexibility and adaptive behavior is necessary and can be a source of success ("work as done").

SAMPLE ASRS EVENT 1347601

ASRS = Aviation Safety Reporting System

LIMITATIONS OF ASRS RECORDS

- Self-reported, subjective, and written from memory
- May be incomplete or biased, based on a single perspective of the event
- Generally filed because an undesirable outcome occurred or almost occurred. Not filed when all goes well.
  - Harder to evaluate Safety 2 in ASRS events.
- Cannot make any statements about rates of events in general
- Timing issues
  - May refer to out-of-date arrivals, departures, or approaches.
  - Old charts may not be available.
  - Delayed entry of events due to processing time (a few months).

ACCESSION NUMBER (ACN) 1347601

- Obtained from NASA public database, Aviation Safety Reporting System (ASRS) https://asrs.arc.nasa.gov/search/database.html
- Coded fields include time/day (201604 = April 2016) and other facts about the submitted report.
- Narrative has been de-identified.
- Some acronyms

  FO = first officer

  DFW = Dallas-Fort Worth airport in Texas (TX)

  DAL = Dallas Love Field, TX

  FL### = Flight Level (altitude in thousands of feet), e.g., FL340 = 34,000 ft altitude

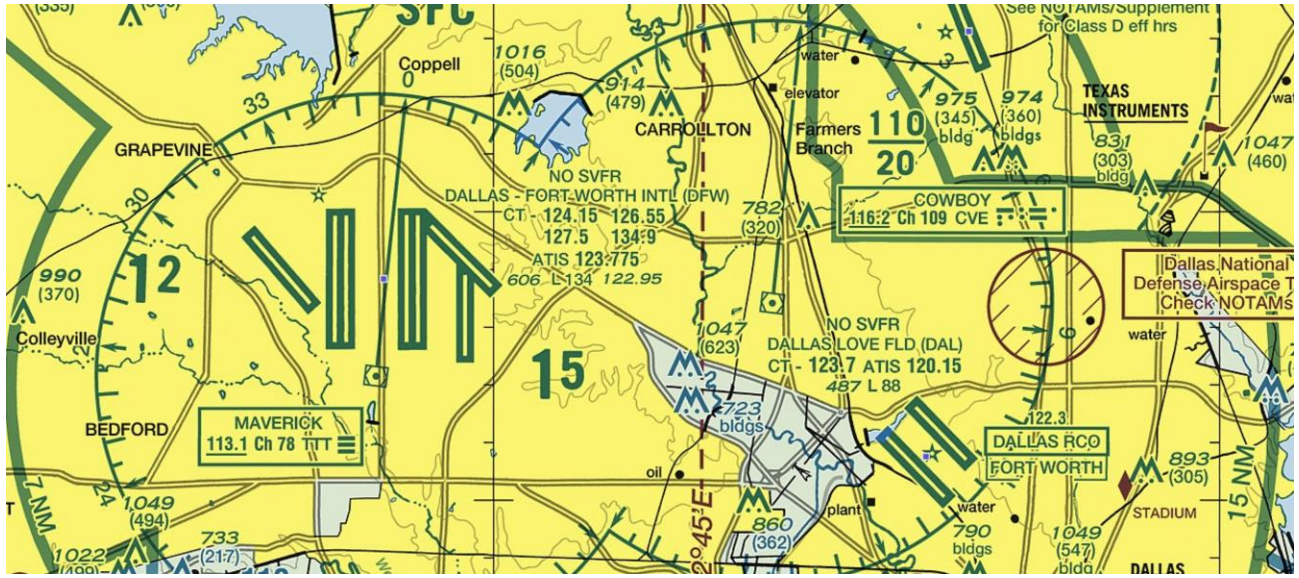
  ACARS = text messaging system for airline company communications

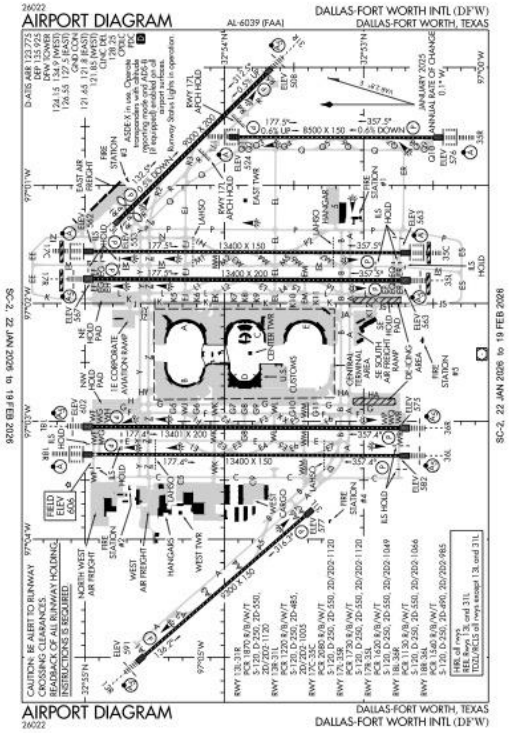
  ASAP = airlines' internal safety reporting system for pilots

  MFD = multi-function display in the flight deck

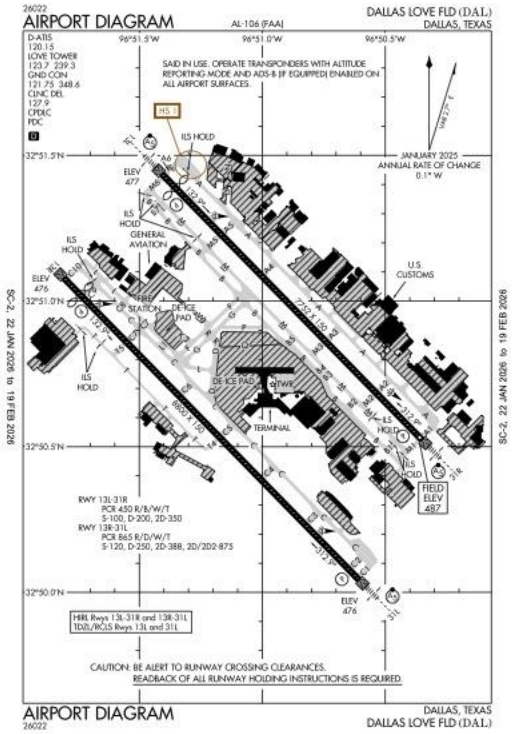
  PA = public announcement

DALLAS-FORT WORTH (DFW) AND DALLAS (DAL)





DFW



DAL

Note that both airports have a “13L” runway, of similar lengths. The distance between these airports is about 10 miles.

Chart images from airnav.com

EXAMPLE ASRS EVENT

- What happened? Summarize in your own words.
  - What was the sequence of events? (facts)
  - What were the “threats” in this situation?
    (See list on earlier slide, Operational Complexity, for ideas)
- What information could you find online that would help you understand this event?
- How did the flight crew contribute to safety?
  - What might have happened with a different crew?
  - What might have happened if there were no crew, but an autonomous system instead?

EXTRA

Connection to earlier reading

LIMITS OF EXPERTISE

- Why do highly skilled pilots make errors?
  - Important to avoid hindsight bias
  - Examine the situation as the pilots understood it at the time to understand their responses
- Need to understand skilled performance and factors that affect the probability of errors
  - Human cognitive and perceptual limitations
  - Task demands
  - Events in the environment
  - Social and organizational factors